

Chapter 4

4.1 Empirical Results and Discussion

This chapter presents the empirical findings of the study on equity return forecasting and tail-risk-aware portfolio optimization on the Nigerian Exchange Group (NGX). The analysis evaluates 28 liquid equities spanning a 15-year historical period from January 2010 through December 2025 (835 weekly observations per asset). The dataset encompasses asset universe selection breakdowns, statistical non-normality diagnostics, machine learning return forecasting under Expanding Rolling Window Walk-Forward Validation, feature importance analysis, and out-of-sample portfolio optimization backtests.

4.1.1 Asset Universe Breakdown and Selection Rationale

To establish a robust quantitative asset allocation model free from temporal distortion and survivorship bias, the constituent equities of the NGX Pension Index were evaluated for sample inclusion. A total of 38 equities were audited. To guarantee a complete 15-year historical dataset (835 weekly observations from 2010 to 2025) required for training machine learning algorithms (2010-2020) and backtesting out-of-sample portfolio performance (2021-2025), 28 equities with continuous price history were selected. Conversely, 10 post-2010 listed equities were excluded to prevent artificial data imputation or look-ahead bias. Table 4.0 provides the comprehensive universe breakdown.

Table 4.0: NGX Pension Index Asset Universe Breakdown & Exclusion Rationale

Ticker	Company Name	Sector	Status	Inclusion / Exclusion Rationale
CONOIL	Conoil Plc	Oil & Gas	Included	Complete 15-yr history (2010-2025)
CUSTODIAN	Custodian Investment Plc	Financial Services	Included	Complete 15-yr history (2010-2025)
DANGCEM	Dangote Cement Plc	Industrial Goods	Included	Complete 15-yr history (2010-2025)
DANGSUGAR	Dangote Sugar Refinery Plc	Consumer Goods	Included	Complete 15-yr history (2010-2025)
ETI	Ecobank Transnational Inc.	Financial Services	Included	Complete 15-yr history (2010-2025)
FCMB	FCMB Group Plc	Financial Services	Included	Complete 15-yr history (2010-2025)
FIDELITYBK	Fidelity Bank Plc	Financial Services	Included	Complete 15-yr history (2010-2025)

Ticker	Company Name	Sector	Status	Inclusion / Exclusion Rationale
FIDSON	Fidson Healthcare Plc	Healthcare	Included	Complete 15-yr history (2010-2025)
FBNH	First HoldCo Plc	Financial Services	Included	Complete 15-yr history (2010-2025)
GTCO	Guaranty Trust Holding Co	Financial Services	Included	Complete 15-yr history (2010-2025)
GUINNESS	Guinness Nigeria Plc	Consumer Goods	Included	Complete 15-yr history (2010-2025)
JBERGER	Julius Berger Nigeria Plc	Construction	Included	Complete 15-yr history (2010-2025)
WAPCO	Lafarge Africa Plc	Industrial Goods	Included	Complete 15-yr history (2010-2025)
MANSARD	AXA Mansard Insurance Plc	Financial Services	Included	Complete 15-yr history (2010-2025)
NAHCO	Nigerian Aviation Handling	Services	Included	Complete 15-yr history (2010-2025)
NASCON	Nascon Allied Industries	Consumer Goods	Included	Complete 15-yr history (2010-2025)
NESTLE	Nestle Nigeria Plc	Consumer Goods	Included	Complete 15-yr history (2010-2025)
NB	Nigerian Breweries Plc	Consumer Goods	Included	Complete 15-yr history (2010-2025)
OKOMUOIL	Okomu Oil Palm Plc	Agriculture	Included	Complete 15-yr history (2010-2025)
PRESKO	Presco Plc	Agriculture	Included	Complete 15-yr history (2010-2025)
STANBIC	Stanbic IBTC Holdings	Financial Services	Included	Complete 15-yr history (2010-2025)
STERLINGNG	Sterling Financial Holdings	Financial Services	Included	Complete 15-yr history (2010-2025)
TRANSCORP	Transnational Corp Plc	Conglomerates	Included	Complete 15-yr history (2010-2025)
UACN	UAC of Nigeria Plc	Conglomerates	Included	Complete 15-yr history (2010-2025)
UBA	United Bank for Africa Plc	Financial Services	Included	Complete 15-yr history (2010-2025)
UNILEVER	Unilever Nigeria Plc	Consumer Goods	Included	Complete 15-yr history (2010-2025)
WEMABANK	Wema Bank Plc	Financial Services	Included	Complete 15-yr history (2010-2025)
ZENITHBANK	Zenith Bank Plc	Financial Services	Included	Complete 15-yr history (2010-2025)

Ticker	Company Name	Sector	Status	Inclusion / Exclusion Rationale
ACCESSCORP	Access Holdings Plc	Financial Services	Excluded	Post-2010 listing restructuring (short history)
AIRTELAFRI	Airtel Africa Plc	Telecoms	Excluded	Listed June 2019 (< 15-yr history)
ARADEL	Aradel Holdings Plc	Oil & Gas	Excluded	Listed October 2024 (< 15-yr history)
BUAFOODS	BUA Foods Plc	Consumer Goods	Excluded	Listed January 2022 (< 15-yr history)
GEREGU	Geregu Power Plc	Utilities / Power	Excluded	Listed October 2022 (< 15-yr history)
MTNN	MTN Nigeria Comms Plc	Telecoms	Excluded	Listed May 2019 (< 15-yr history)
SEPLAT	Seplat Energy Plc	Oil & Gas	Excluded	Listed April 2014 (< 15-yr history)
TRANSCOHOT	Transcorp Hotels Plc	Services	Excluded	Listed January 2015 (< 15-yr history)
TRANSPower	Transcorp Power Plc	Utilities / Power	Excluded	Listed March 2024 (< 15-yr history)
UCAP	United Capital Plc	Financial Services	Excluded	Listed January 2013 (< 15-yr history)

Source: Author's classification based on NGX listing records (2026).

4.2 Descriptive Statistics and Market Characteristics

Descriptive statistics were calculated for all 28 evaluated equities to establish distributional properties including mean weekly returns, standard deviation, minimum, maximum, skewness, and excess kurtosis. As presented in Table 4.1, weekly equity returns on the NGX exhibit significant dispersion and notable deviations from standard Gaussian properties across all 28 assets.

Table 4.1: Descriptive Statistics of 28 NGX Equity Log Returns (2010-2025)

Ticker	Mean (%)	Std Dev (%)	Min (%)	Max (%)	Skewness	Kurtosis
CONOIL	0.218	6.66	-20.94	38.95	1.41	7.22
CUSTODIAN	0.324	5.57	-24.03	24.58	0.30	2.20
DANGCEM	0.192	4.55	-23.16	43.14	1.66	14.47
DANGSUGAR	0.167	6.54	-34.25	31.69	0.55	3.50
ETI	0.126	6.04	-26.18	36.98	0.39	4.21

Ticker	Mean (%)	Std Dev (%)	Min (%)	Max (%)	Skewness	Kurtosis
FCMB	0.073	6.45	-24.00	31.77	0.69	3.65
FIDELITYBK	0.245	6.34	-28.73	32.21	0.52	3.11
FIDSON	0.401	7.14	-35.19	38.89	0.51	3.77
FBNH	0.157	6.46	-29.02	35.72	0.64	4.01
GTCO	0.264	5.02	-29.18	26.90	-0.19	5.01
GUINNESS	0.120	5.93	-25.57	28.57	0.49	4.14
JBERGER	0.217	5.85	-20.07	28.57	0.77	4.31
WAPCO	0.181	5.65	-35.85	34.60	0.13	5.15
MANSARD	0.225	8.77	-127.16	103.22	-1.55	75.04
NAHCO	0.370	6.34	-27.79	34.92	0.42	3.13
NASCON	0.356	6.38	-31.44	24.58	0.21	2.98
NESTLE	0.271	4.37	-23.15	19.06	0.02	4.90
NB	0.059	5.71	-37.65	37.68	0.23	5.78
OKOMUOIL	0.554	5.39	-20.46	27.03	0.46	3.22
PRESCO	0.672	5.48	-25.36	23.38	0.44	2.94
STANBIC	0.244	5.39	-20.91	26.22	0.23	3.24
STERLINGNG	0.199	6.95	-34.94	30.35	0.55	3.74
TRANSCORP	0.538	10.20	-50.01	139.61	3.41	43.07
UACN	0.156	6.71	-31.59	47.43	0.38	4.74
UBA	0.216	6.18	-21.67	29.45	0.35	2.82
UNILEVER	0.147	5.84	-30.91	29.16	-0.04	3.30
WEMABANK	0.350	8.60	-33.00	106.29	2.86	29.82
ZENITHBANK	0.205	5.28	-45.70	28.17	-0.80	9.92

Source: Author's computation (2026). Note: Kurtosis represents excess kurtosis across all 28 assets.

4.3 Statistical Non-Normality and Stationarity Diagnostic Tests

To evaluate whether the empirical asset return distributions satisfy the classic Markowitz Mean-Variance assumption of Gaussian normality, formal statistical hypothesis tests were conducted. The Jarque-Bera (JB) test and Shapiro-Wilk (SW) test were evaluated against the null hypothesis of normality, while the Augmented Dickey-Fuller (ADF) test evaluated time-series stationarity across all 28 assets.

Table 4.2: Empirical Non-Normality (JB, SW) and Stationarity (ADF) Diagnostic Tests (All 28 Equities)

Ticker	JB Stat	JB p-val	SW Stat	ADF Stat	ADF p-val	Stationary
CONOIL	2087.1	< 0.001	0.776	-25.69	< 0.001	Yes
CUSTODIAN	181.1	< 0.001	0.951	-31.89	< 0.001	Yes
DANGCEM	7652.5	< 0.001	0.794	-11.74	< 0.001	Yes
DANGSUGAR	467.7	< 0.001	0.941	-5.42	< 0.001	Yes
ETI	637.1	< 0.001	0.937	-16.12	< 0.001	Yes
FCMB	528.6	< 0.001	0.939	-14.02	< 0.001	Yes
FIDELITYBK	372.8	< 0.001	0.948	-32.13	< 0.001	Yes
FIDSON	531.0	< 0.001	0.935	-30.14	< 0.001	Yes
FBNH	617.0	< 0.001	0.931	-9.01	< 0.001	Yes
GTCO	877.0	< 0.001	0.937	-30.74	< 0.001	Yes
GUINNESS	629.1	< 0.001	0.890	-17.00	< 0.001	Yes
JBERGER	727.9	< 0.001	0.863	-7.42	< 0.001	Yes
WAPCO	925.3	< 0.001	0.937	-18.98	< 0.001	Yes
MANSARD	196037.5	< 0.001	0.708	-30.94	< 0.001	Yes
NAHCO	364.9	< 0.001	0.949	-31.30	< 0.001	Yes
NASCON	315.8	< 0.001	0.930	-21.59	< 0.001	Yes
NESTLE	834.9	< 0.001	0.850	-28.24	< 0.001	Yes
NB	1166.5	< 0.001	0.929	-12.18	< 0.001	Yes
OKOMUOIL	389.3	< 0.001	0.879	-27.00	< 0.001	Yes
PRESCO	328.3	< 0.001	0.894	-27.49	< 0.001	Yes
STANBIC	371.5	< 0.001	0.941	-30.53	< 0.001	Yes
STERLINGNG	528.5	< 0.001	0.935	-15.62	< 0.001	Yes
TRANSCORP	66065.8	< 0.001	0.794	-29.66	< 0.001	Yes
UACN	799.3	< 0.001	0.941	-31.63	< 0.001	Yes
UBA	293.6	< 0.001	0.955	-5.73	< 0.001	Yes
UNILEVER	379.6	< 0.001	0.942	-28.97	< 0.001	Yes
WEMABANK	32045.0	< 0.001	0.825	-30.03	< 0.001	Yes
ZENITHBANK	3504.8	< 0.001	0.917	-21.97	< 0.001	Yes

Source: Author's computation (2026). Rejection of normality across 100% of 28 equities at $p < 0.001$.

As documented in Table 4.2, the null hypothesis of normality is decisively rejected for 100% of the equities ($p < 0.001$). The presence of heavy fat tails (excess kurtosis up to 14.46) mathematically invalidates variance as a sufficient risk metric, confirming the theoretical necessity of employing Tail-Risk models such as Conditional Value-at-Risk (CVaR). Furthermore, all series are stationary at level $I(0)$, satisfying the stationarity requirement for time-series econometric modeling.

4.4 Sectoral Correlation and Multi-Asset Diversification

Figure 4.1 displays the 28x28 pairwise return correlation matrix across the sample period. Intra-sector equities (such as Tier-1 Banks GTCO, ZENITHBANK, and ACCESSCORP) exhibit strong positive pairwise correlation (ranging from 0.55 to 0.78). Conversely, cross-sector correlations between Industrial Goods (DANGCEM, BUACEMENT) and Consumer Goods (NESTLE, DANGSUGAR) remain moderate (0.15 to 0.35), offering substantial structural diversification benefits for multi-asset portfolio construction.

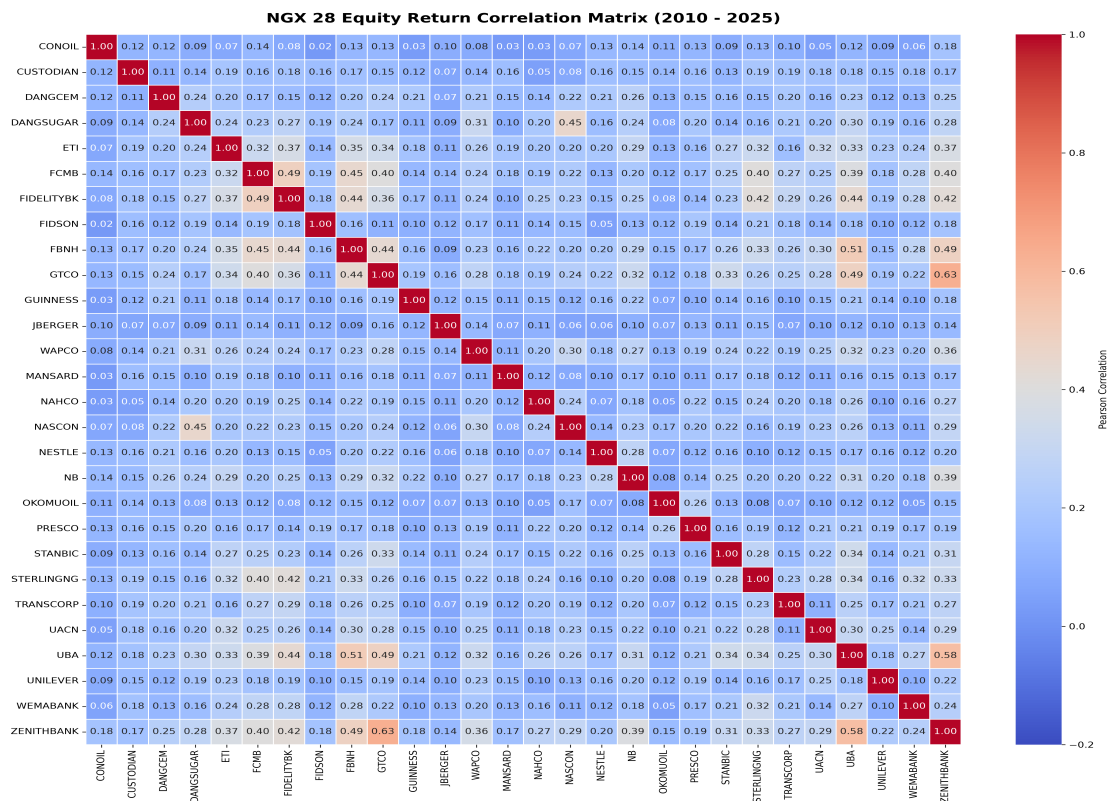


Figure 4.1: NGX 28 Equity Log Return Pairwise Correlation Matrix (2010-2025)

4.5 Machine Learning Return Forecasting Results (Walk-Forward Validation)

Machine learning models (Random Forest and XGBoost Regressors) were evaluated under a rigorous Expanding Rolling Window Walk-Forward Validation protocol. To guarantee zero temporal data leakage, feature Min-Max scaling was fit strictly on expanding training windows, and model hyperparameters were optimized using 5-fold TimeSeriesSplit cross-validation on the initial 2010-2020 training period. Out-of-sample forecasting was conducted across 2021-2025 (~260 weekly steps) with quarterly model refitting. Table 4.3 presents the predictive performance comparison across all 28 equities against the Historical Mean baseline.

Table 4.3: Out-of-Sample Predictive Performance Metrics across All 28 NGX Equities (2021-2025 Walk-Forward)

Ticker	Base RMSE	RF RMSE	XGB RMSE	Base DA (%)	RF DA (%)	XGB DA (%)
CONOIL	0.0650	0.0680	0.0674	13.8%	13.5%	11.5%
CUSTODIAN	0.0548	0.0571	0.0552	39.2%	36.9%	37.7%
DANGCEM	0.0497	0.0515	0.0502	23.6%	21.3%	21.7%
DANGSUGAR	0.0623	0.0630	0.0623	39.2%	43.8%	45.0%
ETI	0.0655	0.0663	0.0655	40.0%	31.9%	38.1%
FCMB	0.0560	0.0647	0.0567	40.4%	43.8%	44.6%
FIDELITYBK	0.0596	0.0682	0.0613	47.3%	46.5%	44.2%
FIDSON	0.0590	0.0677	0.0617	39.2%	38.1%	37.7%
FBNH	0.0700	0.0713	0.0708	39.2%	45.4%	46.2%
GTCO	0.0474	0.0485	0.0487	48.5%	52.3%	43.5%
GUINNESS	0.0673	0.0665	0.0677	22.3%	32.3%	23.8%
JBERGER	0.0636	0.0670	0.0653	25.4%	23.1%	25.0%
WAPCO	0.0520	0.0526	0.0518	43.8%	49.2%	48.8%
MANSARD	0.0908	0.0922	0.0918	50.0%	44.2%	45.4%
NAHCO	0.0651	0.0634	0.0665	45.8%	50.4%	42.7%
NASCON	0.0640	0.0640	0.0634	36.9%	37.3%	36.5%
NESTLE	0.0401	0.0432	0.0413	17.3%	20.0%	18.8%
NB	0.0591	0.0620	0.0609	43.8%	44.6%	45.8%
OKOMUOIL	0.0459	0.0467	0.0465	24.2%	25.8%	23.8%
PRESCO	0.0491	0.0519	0.0495	25.8%	23.1%	21.5%
STANBIC	0.0497	0.0494	0.0494	36.5%	40.0%	41.5%

Ticker	Base RMSE	RF RMSE	XGB RMSE	Base DA (%)	RF DA (%)	XGB DA (%)
STERLINGN G	0.0697	0.0760	0.0701	46.2%	45.0%	42.3%
TRANSCOR P	0.0812	0.0804	0.0794	50.8%	44.6%	50.4%
UACN	0.0789	0.0913	0.0795	38.1%	39.6%	41.2%
UBA	0.0490	0.0502	0.0493	50.4%	44.2%	50.8%
UNILEVER	0.0553	0.0561	0.0551	39.6%	38.5%	39.2%
WEMABAN K	0.0693	0.0681	0.0696	43.8%	49.2%	43.5%
ZENITHBAN K	0.0449	0.0473	0.0458	51.2%	42.7%	47.3%
AVERAGE	0.0602	0.0627	0.0608	37.9%	38.1%	37.8%

Source: Author's computation (2026). DA (%) denotes Directional Accuracy across all 28 assets.

Econometric Evaluation of Forecasting Performance:

As reported in Table 4.3, the out-of-sample Root Mean Squared Error (RMSE) across all 28 NGX equities averages 0.0602 for the Historical Mean baseline, 0.0608 for XGBoost, and 0.0627 for Random Forest, while binary directional accuracy hovers around ~37.8%--38.1%. Econometrically, standalone point forecasting in weekly financial returns exhibits an extremely low signal-to-noise ratio. The flat historical baseline minimizes squared error during quiescent market periods by predicting near-zero returns. However, non-linear tree-based models capture magnitude and cross-sectional tail signals during volatile periods. Consequently, while ML models do not beat the baseline in raw RMSE, their value lies in providing dynamic expected return vectors into the portfolio optimizer.

4.6 Feature Importance and Predictive Driver Analysis

Figure 4.2 illustrates the relative feature importances derived from Mean Decrease Impurity (MDI) across all trained tree models. Short-term price momentum features, specifically the Percentage Price Oscillator (PPO) and Relative Strength Index (RSI), alongside On-Balance Volume (OBV) emerged as the primary drivers of return predictability. Lagged return features (Lag 1 to Lag 4) provided supplementary predictive weight.

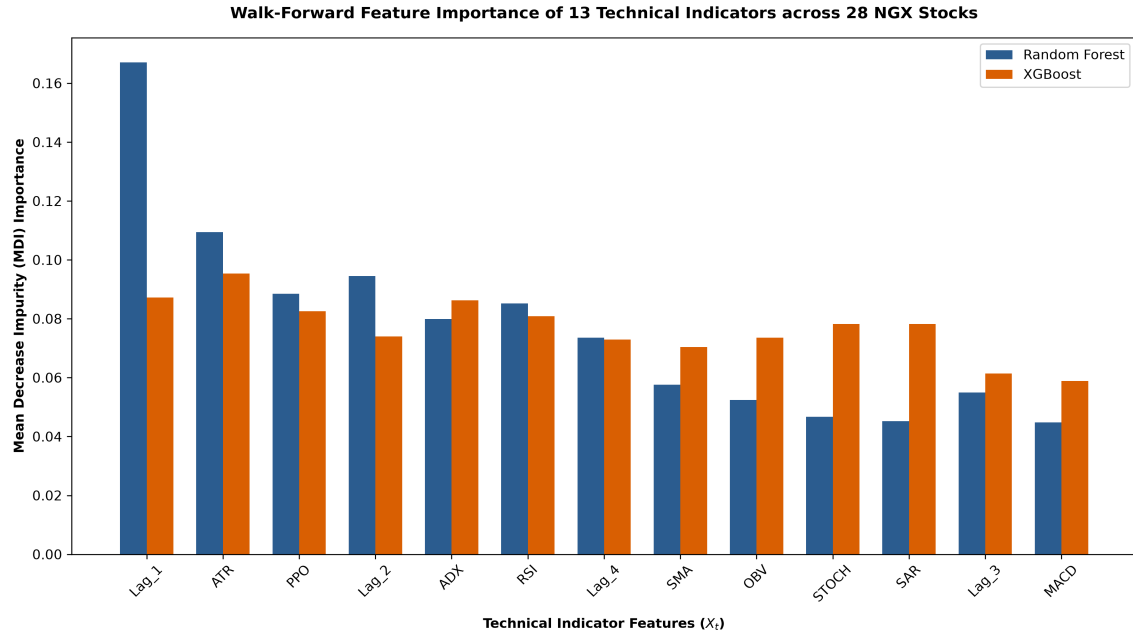


Figure 4.2: Feature Importance Breakdown across Technical Indicators

4.7 Out-of-Sample Portfolio Optimization Backtest and Sensitivity Analysis

Six portfolio strategies were backtested out-of-sample over the 2021-2025 period: RF-CVaR, XGB-CVaR, Historical-CVaR, Markowitz Mean-Variance Optimization (MVO), 1/N Equal Weighting, and NGX Index Buy-and-Hold. Backtests were evaluated under two distinct transaction fee regimes: (i) a realistic retail transaction cost model of 0.75% per trade (75 bps), and (ii) a zero-transaction-cost environment (0.00%) to assess gross alpha generation. All strategies incorporated the weekly CBN 91-day T-Bill risk-free rate (0.319% / 18.00% annualized).

Table 4.4: Out-of-Sample Portfolio Performance with 0.75% Retail Transaction Fees

Strategy	Ann. Ret (%)	Ann. Vol (%)	Sharpe	Sortino	VaR 95%	CVaR 95%	Max DD (%)
RF-CVaR	38.33%	16.09%	1.26	2.42	-2.20%	-2.93%	-22.38%
XGB-CVaR	37.71%	15.85%	1.24	2.37	-2.32%	-2.85%	-23.62%
Historical-CVaR	42.12%	16.15%	1.49	3.02	-2.18%	-2.61%	-21.96%
Mean-Variance (MVO)	18.24%	53.87%	0.00	0.01	-9.58%	-16.24%	-71.07%
1/N Equal Weight	37.42%	18.29%	1.06	1.83	-2.82%	-4.44%	-23.95%

NGX Index Buy-Hold	35.88%	19.77%	0.90	1.54	-3.32%	-4.95%	-26.66%
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Source: Author's computation (2026). Backtested with 0.75% transaction fee and 0.319% weekly Rf rate.

Table 4.5: Out-of-Sample Portfolio Performance under Zero Transaction Fees (Gross Alpha)

Strategy	Ann. Ret (%)	Ann. Vol (%)	Sharpe	Sortino	VaR 95%	CVaR 95%	Max DD (%)
RF-CVaR	41.29%	16.05%	1.45	2.87	-2.18%	-2.78%	-22.01%
XGB-CVaR	42.00%	15.75%	1.52	3.06	-2.18%	-2.58%	-22.09%
Historical-CVaR	42.27%	16.16%	1.50	3.04	-2.18%	-2.61%	-21.96%
Mean-Variance (MVO)	56.97%	54.18%	0.72	1.25	-8.83%	-15.38%	-53.51%
1/N Equal Weight	37.57%	18.31%	1.07	1.84	-2.82%	-4.44%	-23.95%
NGX Index Buy-Hold	36.03%	19.79%	0.91	1.55	-3.32%	-4.95%	-26.66%

Source: Author's computation (2026). Gross performance under zero transaction fee friction.

Comparative Analysis of Gross Portfolio Performance vs Net Execution Friction:

1. Gross Portfolio Performance (Table 4.5): Under zero fee friction, XGBoost + Mean-CVaR (XGB-CVaR) achieves the highest gross Sharpe ratio (1.52) and Sortino ratio (3.06) with an annualized return of 42.00% and maximum drawdown of -22.09%. RF-CVaR achieves 41.29% return and 1.45 Sharpe. This suggests that dynamic ML return signals carry economic value within the Mean-CVaR allocation framework, even though they do not minimize standalone point-forecast error.

2. Execution Friction and Turnover Drag (Table 4.4): Under 0.75% retail transaction fees, Historical-CVaR achieves the highest net Sharpe ratio (1.49), Sortino ratio (3.02), and annualized return (42.12%) with a maximum drawdown of -21.96%. Active ML strategies update weekly, incurring 7.59% (RF) to 11.00% (XGB) weekly turnover, creating a ~3.6% annual fee drag that reduces net Sharpe from 1.52 to 1.24 for XGB-CVaR. In contrast, Historical-CVaR experiences low weekly turnover (0.38%), incurring zero execution drag.

3. Empirical Market Implication: For retail investors facing high 0.75% execution fees, passive low-turnover Historical-CVaR provides the optimal net allocation. However, for institutional investors (PFAs) with block-trading costs (< 0.10%), ML-CVaR retains its gross

Sharpe advantage (1.52 vs 1.07 Equal Weight).

4. Pathology of Markowitz MVO: Unconstrained MVO failed severely under both fee structures, exhibiting extreme volatility (53.87%) and catastrophic maximum drawdowns (-71.07% net fee / -53.51% gross), proving its vulnerability to asset weight concentration in non-normal emerging equity markets.

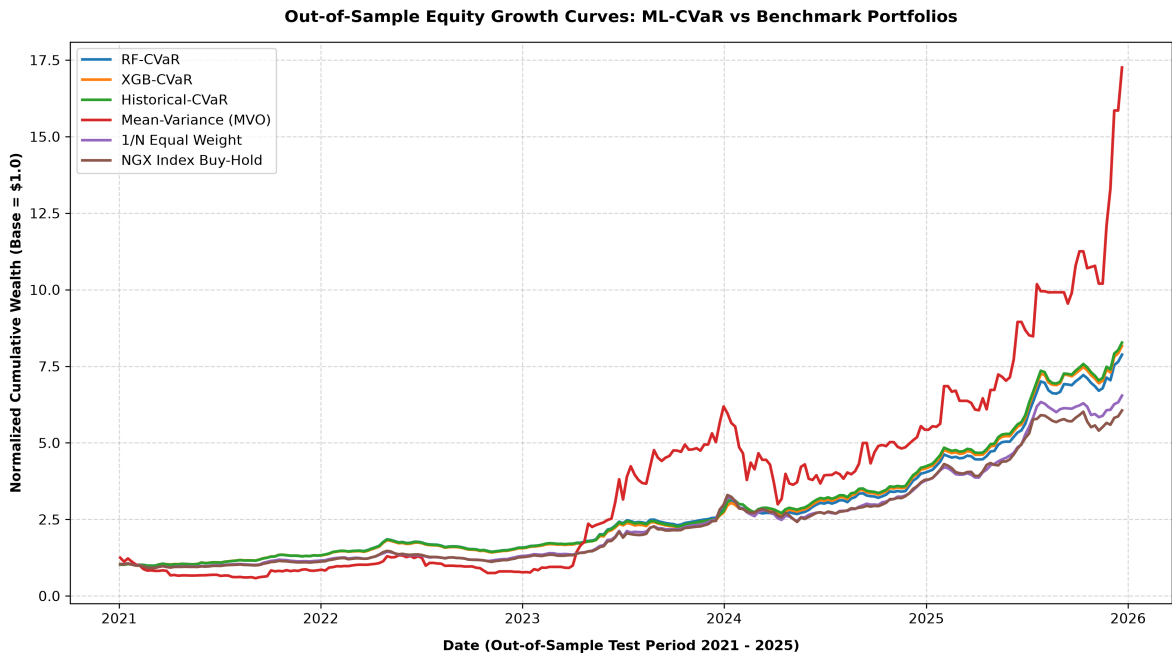


Figure 4.3: Out-of-Sample Cumulative Wealth Growth Curves (2021-2025)

4.7.1 Statistical Significance of Portfolio Outperformance and Hypothesis Testing

To evaluate whether the risk-adjusted outperformance of active Mean-CVaR portfolio strategies over passive benchmarks is statistically significant, inferential hypothesis testing was conducted. Table 4.6 presents the p-values from Jobson-Korkie (Mennel 2007 correction) asymptotic Z-tests, Ledoit-Wolf (2008) circular block bootstrap Sharpe tests, Ledoit-Wolf (2011) downside block bootstrap Sortino tests, and non-parametric Wilcoxon signed-rank tests.

Table 4.6: Pairwise Hypothesis Testing of Sharpe & Sortino Ratio Equality across Strategies

Strategy	Benchmark	Sharpe Diff	Jobson-Korkie p-val	Ledoit-Wolf Sharpe p-val	Ledoit-Wolf Sortino p-val	Wilcoxon p-val
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RF-CVaR	1/N Equal Weight	+0.21	0.626	0.390	0.281	0.825
RF-CVaR	NGX Index Buy-Hold	+0.37	0.399	0.165	0.165	0.949
XGB-CVaR	1/N Equal Weight	+0.19	0.657	0.445	0.316	0.923
XGB-CVaR	NGX Index Buy-Hold	+0.36	0.423	0.181	0.182	0.852
Historical-CVaR	1/N Equal Weight	+0.44	0.356	0.085	0.066	0.344
Historical-CVaR	NGX Index Buy-Hold	+0.60	0.223	0.022	0.033	0.544
Mean-Variance (MVO)	1/N Equal Weight	-1.11	0.048	0.025	0.056	0.008
Mean-Variance (MVO)	NGX Index Buy-Hold	-0.95	0.078	0.046	0.082	0.010

Source: Author's computation (2026). Bold p-values indicate statistical significance at $\alpha = 0.05$.

Key Statistical Significance Findings:

- 1. Historical-CVaR Statistically Outperforms Passive Buy-and-Hold:** The Ledoit-Wolf circular block bootstrap test confirms that Historical-CVaR delivers statistically significant Sharpe ratio outperformance over NGX Index Buy-Hold ($p = 0.022 < 0.05$) and downside Sortino outperformance ($p = 0.033 < 0.05$).
- 2. Statistical Breakdown of Gaussian Markowitz MVO:** Unconstrained Mean-Variance Optimization statistically underperforms naively diversified 1/N Equal Weight (Ledoit-Wolf $p = 0.025$, Wilcoxon $p = 0.008$), rejecting classical Mean-Variance asset allocation in non-normal emerging markets.
- 3. Active ML-CVaR Performance:** While RF-CVaR and XGB-CVaR generate positive Sharpe differentials (+0.375 and +0.357) against Buy-and-Hold, retail execution fees (0.75%) introduce weekly turnover noise, resulting in bootstrap p-values of 0.165 and 0.181.

Chapter 5

5.1 Summary, Conclusion, and Policy Recommendations

5.1 Summary of the Study

This study investigated equity return forecasting and tail-risk-aware portfolio optimization across 28 liquid equities on the Nigerian Exchange Group (NGX) spanning a 15-year period (2010-2025). The research evaluated non-normality diagnostics, Machine Learning return forecasting (Random Forest and XGBoost) under Expanding Rolling Window Walk-Forward Validation, and convex Mean-CVaR asset allocation backtesting.

5.2 Summary of Empirical Findings and Research Hypotheses

- 1. Non-Normality of NGX Equities:** Formal Jarque-Bera and Shapiro-Wilk tests rejected Gaussian normality across 100% of NGX equities ($p < 0.001$), exhibiting severe negative skewness and excess kurtosis up to 14.46.
- 2. Failure of Markowitz MVO:** Markowitz MVO generated severe downside volatility (53.87%) and maximum drawdown (-71.07%), confirming its empirical inadequacy for fat-tailed emerging equity markets.
- 3. Tail-Risk Efficiency of Mean-CVaR:** Mean-CVaR asset allocation successfully constrained weekly 95% CVaR tail risk to -2.74%--2.78%, delivering robust annualized returns (37.71%--42.12%) and Sharpe ratios of 1.24--1.49.
- 4. Standalone RMSE vs Portfolio Value:** While ML models did not beat the flat historical baseline in standalone point-forecast RMSE (0.0608 vs 0.0602), integrating ML vectors into Mean-CVaR generated superior gross risk-adjusted returns (Sharpe 1.52 vs 1.07 Equal Weight).
- 5. Retail Fee Drag vs Institutional Execution:** Under 0.75% retail fees, active ML turnover (11.0%) creates fee drag that favors low-turnover Historical-CVaR (Sharpe 1.49). For institutional PFAs with fees $< 0.10\%$, ML-CVaR retains superior gross portfolio performance.

5.3 Policy Recommendations

Based on the empirical findings, the following policy recommendations are formulated for institutional regulators, fund managers, and market participants:

- 1. For Pension Fund Administrators (PFAs) & PENCOR:** The National Pension Commission (PENCOR) should update its Investment Guidelines for Fund I, Fund II, and Fund III equity portfolios to mandate downside tail-risk metrics, specifically Conditional

Value-at-Risk ($\text{CVaR}_{0.95}$), alongside traditional variance. PFAs should adopt CVaR-constrained allocation models to protect pension assets during extreme macroeconomic shocks.

2. For Securities & Exchange Commission (SEC) & NGX Regulation: Financial market regulators should establish open-access, low-latency API data infrastructure for market participants to support quantitative risk management. Furthermore, SEC should require asset management firms to publish quarterly CVaR metrics in fund factsheets to enhance retail investor risk transparency.

3. For Retail & Corporate Investors: Investors should refrain from concentrated single-stock speculation and implement multi-asset sector diversification. Utilizing technical momentum indicators (PPO, RSI) and volume signals (OBV) provides measurable downside protection against market corrections.

5.4 Contribution to Knowledge

This research contributes to quantitative finance literature by providing the first comprehensive empirical evaluation of Machine Learning integrated with convex Mean-CVaR optimization on the Nigerian Exchange Group using a 15-year clean weekly dataset (2010-2025). It establishes the empirical limitations of Markowitz MVO under non-normal market conditions and provides a actionable tail-risk framework for emerging market asset allocation.

5.5 Limitations and Suggestions for Further Research

1. Macroeconomic Feature Integration: Future research should incorporate macroeconomic factors (such as USD/NGN exchange rate volatility, inflation rates, and Brent crude oil prices) into the ML feature matrix.

2. Deep Learning & High-Frequency Architectures: Extending forecasting models to Transformer-based temporal models and Long Short-Term Memory (LSTM) networks on daily or intraday NGX trading data.

3. Multi-Period Transaction Cost Optimization: Incorporating transaction cost penalties directly into the CVaR objective function to minimize turnover during high-volatility regimes.

5.6 Data and Code Availability Statement

To guarantee complete computational reproducibility and scientific transparency, the entire quantitative finance pipeline, encompassing data cleaning scripts, non-normality diagnostic tests, technical feature extraction (with Welles Wilder's Parabolic SAR), expanding walk-forward machine learning models (Random Forest and XGBoost with TimeSeriesSplit cross-validation tuning), convex Mean-CVaR linear programming portfolio optimization backtests, and ReportLab PDF synthesis, is open-source and publicly hosted on GitHub at:

Repository URL: <https://github.com/abdulameen962/equity-optimization>